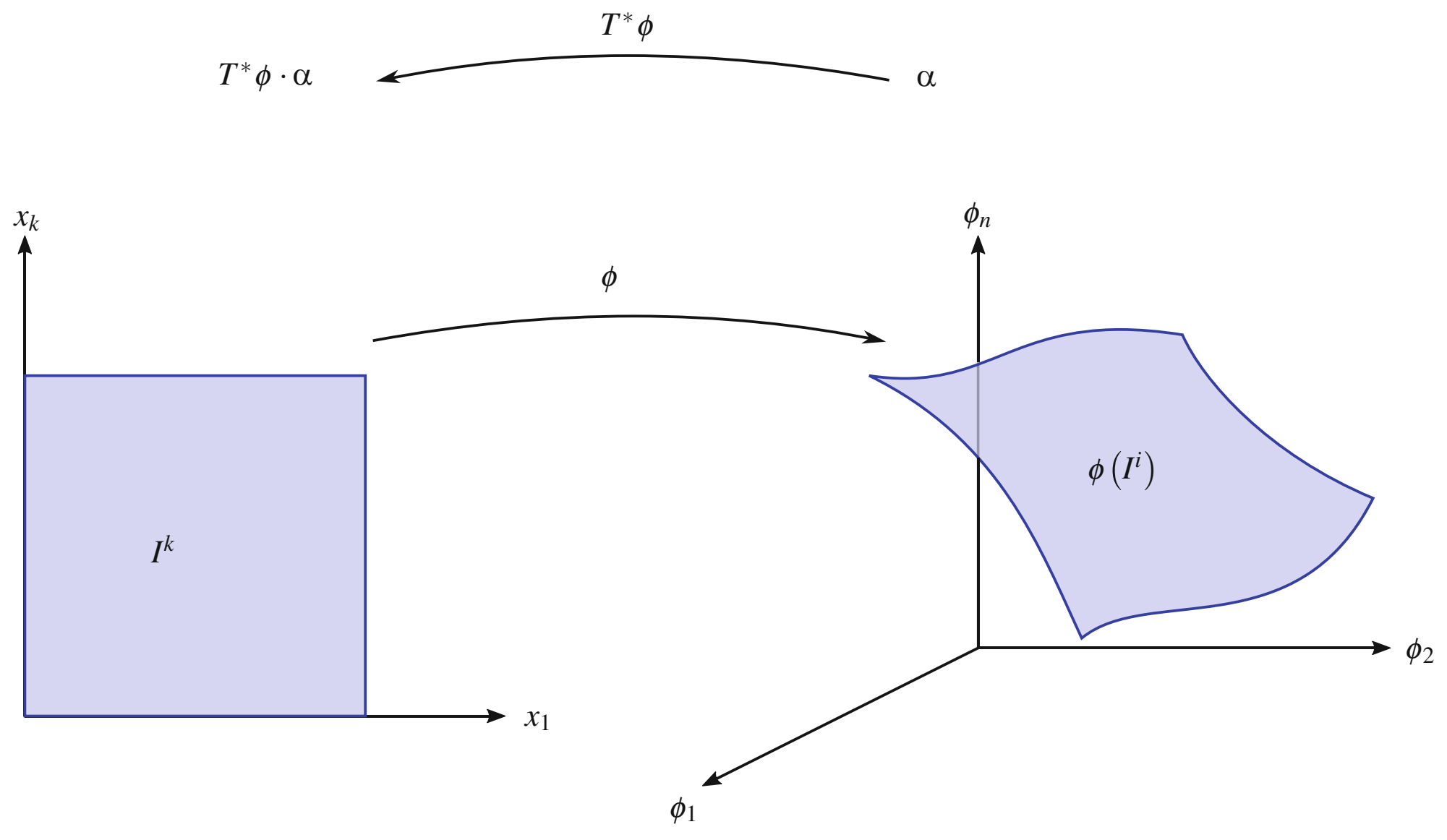

$T^*\phi$
$T^*\phi \cdot \alpha$
α
x_k
ϕ
ϕ_n
I^k
$\phi\left(I^i\right)$
x_1
ϕ_2
ϕ_1